Fallacies of LLM Systems Cheat Sheet

Eight assumptions from ordinary software that break AI in production, and the control that fixes each.

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THE EIGHT FALLACIES

Fallacy	What's actually true	Do this instead
1 · The model is deterministic	Temperature 0 still varies on hosted endpoints: 80 distinct outputs from 1,000 identical requests (Thinking Machines, 2025).	Assert on properties, not strings. Run eval cases several times; report a pass rate. Debug from logged traces, don't re-run.
2 · The output is well-formed	Schema-valid is not correct. Responses get truncated at the token limit or replaced by refusals.	Use structured outputs. Validate shape and meaning. Handle truncation and refusals. Retry once, then fail closed.
3 · The model stays the same	Versions drift (GPT-4 prime test: 84% to 51%) and get retired. Formatting alone moved accuracy up to 76 points.	Pin dated versions. Treat a model change as an upgrade gated on evals. Track deprecation dates.
4 · The demo is the test	Five hand-picked inputs say nothing about the long tail of real traffic.	Eval set from real traces, grown on every failure. CI blocks on it. Check agent paths, not just answers.
5 · Context is free	Every token is billed and slows the first token. Accuracy drops mid-context and as input grows.	Retrieve, don't stuff. Key material at start or end. Measure at your real length. Cache stable prefixes.
6 · The model knows your data	It knows its training data and guesses about yours, because training rewards guessing.	RAG with citations. An explicit "not found" path. Measure faithfulness. Fine-tune for style, not facts.
7 · Input is trusted	Pages, files and tool output can carry instructions (indirect prompt injection, OWASP LLM01).	Least-privilege tools. Human approval for side effects. Keep fetched text out of allow/deny decisions.
8 · Cost and latency are constant	Agents use ~4x the tokens of chat, multi-agent ~15x (Anthropic). Reasoning and retries add more.	Token and step budgets per request. Cost per trace. Watch the worst 1%. Route easy requests to small models.

DESIGN REVIEW: ONE QUESTION PER FALLACY

1. Do tests assert on properties and run each case more than once?
2. What happens on validation failure, truncation or refusal?
3. Is the model version pinned, and do model changes go through the eval gate?
4. Is there an eval set from real traffic that blocks CI?
5. Do we know accuracy and latency at our real context length?
6. Does the system say "not found" when retrieval is empty?
7. Which tools can the model call; which actions need human approval?
8. Is there a per-request token and step budget, with cost tracked per trace?

THE IDEA BEHIND IT

> *The fallacies of distributed computing (Deutsch et al., Sun Microsystems, 1994) start with "the network is reliable".*

- **Reliable network** becomes **deterministic model**
- **Topology doesn't change** becomes **model stays the same**
- **Infinite bandwidth** becomes **free context**
- **Secure network** becomes **trusted input**
- **Zero transport cost** becomes **constant cost**

> **Rule of thumb:** a model call is not a function call. Design it like a call to someone else's service: variable output, changing implementation, metered bill, attack surface.